KOLLAR: UNDER DEPTH AND DIMENSION HYPOTHESES THE PICARD MAP FROM THE PUNCTURED SPECTRUM TO THE PUNCTURED SPECTRUM MOD F IS INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8729-8749 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 8751-8904, the complete author proof (152 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al02 bib-salvage batch). Spans computed from \label (16.5/16.6).

Editorial note. External \cite keys salvaged with card-local bibliography.tex entries copied verbatim from the snapshot my.bib (precedent S3-morph-blow-up-ample-family); URL/ISSN/EPRINT fields dropped as the preamble has no url package.

Editorial note. statement contains reference environment with real content -> patch-sd-envs.tex applied (sd02 lesson).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'proposition-injective-pic' also exists in stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name become stub prereqs; external cites resolved via the card-local bibliography.tex fragment.

Source reference. Original proposition 00L8: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0AAK: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0A80: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:algebraization.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 0F28: sources/stacks-algebraization/pinned-original-source.tex, source lines 8612–8623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0FIW: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0BPY: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0DWZ: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0F20: sources/stacks-algebraization/pinned-original-source.tex, source lines 8904–8904. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0F27: sources/stacks-algebraization/pinned-original-source.tex, source lines 8503–8514. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0F22: sources/stacks-algebraization/pinned-original-source.tex, source lines 8168–8183. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1 (Kollár)**.** [1, Theorem 1.9] *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $f \in \mathfrak{m}$. Assume*

1. *A has a dualizing complex,*
2. *f is a nonzerodivisor,*
3. *$\text{depth}(A/fA) \geq 2$, or equivalently $\text{depth}(A) \geq 3$,*
4. *if $f \in \mathfrak{p} \subset A$ is a prime ideal with $\dim(A/\mathfrak{p}) = 2$, then $\text{depth}(A_\mathfrak{p}) \geq 2$.*

Let U, resp. U_0 be the punctured spectrum of A, resp. A/fA. The map

$$\text{Pic}(U) \to \text{Pic}(U_0)$$

is injective. Finally, if (1), (2), (3), A is (S_2), and $\dim(A) \geq 4$, then (4) holds.

Proof. Let $\mathcal{L}$ be an invertible $\mathcal{O}_U$-module. Observe that $\mathcal{L}$ maps to 0 in $\text{Pic}(U_0)$ if and only if we can extend $\mathcal{L}$ to an invertible coherent triple $(\mathcal{L}, \mathcal{L}_0, \lambda)$ as in Section 0F22. By Proposition 0F27 the function

$$n \longmapsto \chi((\mathcal{L}, \mathcal{L}_0, \lambda)^{\otimes n})$$

is a polynomial P. By Lemma 0F28 we have $P(n) \geq 0$ for all $n \in \mathbf{Z}$ with equality if and only if $\mathcal{L}^{\otimes n}$ is trivial. In particular $P(0) = 0$ and P is either identically zero and we win or P has even degree ≥ 2.

Set $M = \Gamma(U, \mathcal{L})$ and $M_0 = \Gamma(X_0, \mathcal{L}_0) = \Gamma(U_0, \mathcal{L}_0)$. Then M is a finite A-module of depth ≥ 2 and $M_0 \cong A/fA$, see proof of Lemma 0F28. Note that $H^2_\mathfrak{m}(M)$ is

finite A-module by Local Cohomology, Lemma 0BPY and the fact that $H^i_{\mathfrak{m}}(A) = 0$ for $i = 0, 1, 2$ since $\text{depth}(A) \geq 3$. Consider the short exact sequence

$$0 \to M/fM \to M_0 \to Q \to 0$$

Lemma 0F28 tells us Q has finite length equal to $\chi(\mathcal{L}, \mathcal{L}_0, \lambda)$. We obtain $Q = H^1_{\mathfrak{m}}(M/fM)$ and $H^i_{\mathfrak{m}}(M/fM) = H^i_{\mathfrak{m}}(M_0) \cong H^i_{\mathfrak{m}}(A/fA)$ for $i > 1$ from the long exact sequence of local cohomology associated to the displayed short exact sequence. Consider the long exact sequence of local cohomology associated to the sequence $0 \to M \to M \to M/fM \to 0$. It starts with

$$0 \to Q \to H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(A/fA)$$

Using additivity of lengths we see that $\chi(\mathcal{L}, \mathcal{L}_0, \lambda)$ is equal to the length of the image of $H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(A/fA)$.

Let prove the lemma in a special case to elucidate the rest of the proof. Namely, assume for a moment that $H^2_{\mathfrak{m}}(A/fA)$ is a finite length module. Then we would have $P(1) \leq \text{length}_A H^2_{\mathfrak{m}}(A/fA)$. The exact same argument applied to $\mathcal{L}^{\otimes n}$ shows that $P(n) \leq \text{length}_A H^2_{\mathfrak{m}}(A/fA)$ for all n. Thus P cannot have positive degree and we win. In the rest of the proof we will modify this argument to give a linear upper bound for $P(n)$ which suffices.

Let us study the map $H^2_{\mathfrak{m}}(M) \to H^2_{\mathfrak{m}}(M_0) \cong H^2_{\mathfrak{m}}(A/fA)$. Choose a normalized dualizing complex $\omega_A^\bullet$ for A. By local duality (Dualizing Complexes, Lemma 0AAK) this map is Matlis dual to the map

$$\text{Ext}^{-2}_A(M, \omega_A^\bullet) \longleftarrow \text{Ext}^{-2}_A(M_0, \omega_A^\bullet)$$

whose image therefore has the same (finite) length. The support (if nonempty) of the finite A-module $\text{Ext}^{-2}_A(M_0, \omega_A^\bullet)$ consists of $\mathfrak{m}$ and a finite number of primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ containing f with $\dim(A/\mathfrak{p}_i) = 1$. Namely, by Local Cohomology, Lemma 0DWZ the support is contained in the set of primes $\mathfrak{p} \subset A$ with $\text{depth}_{A_\mathfrak{p}}(M_{0,\mathfrak{p}}) + \dim(A/\mathfrak{p}) \leq 2$. Thus it suffices to show there is no prime $\mathfrak{p}$ containing f with $\dim(A/\mathfrak{p}) = 2$ and $\text{depth}_{A_\mathfrak{p}}(M_{0,\mathfrak{p}}) = 0$. However, because $M_{0,\mathfrak{p}} \cong (A/fA)_\mathfrak{p}$ this would give $\text{depth}(A_\mathfrak{p}) = 1$ which contradicts assumption (4). Choose a section $t \in \Gamma(U, \mathcal{L}^{\otimes -1})$ which does not vanish in the points $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$, see Properties, Lemma 0F20. Multiplication by t on global sections determines a map $t : M \to A$ which defines an isomorphism $M_{\mathfrak{p}_i} \to A_{\mathfrak{p}_i}$ for $i = 1, \ldots, r$. Denote $t_0 = t|_{U_0}$ the corresponding section of $\Gamma(U_0, \mathcal{L}_0^{\otimes -1})$ which similarly determines a map $t_0 : M_0 \to A/fA$ compatible with t. We conclude that there is a commutative diagram

$$\begin{array}{ccc} \text{Ext}^{-2}_A(M, \omega_A^\bullet) & \longleftarrow & \text{Ext}^{-2}_A(M_0, \omega_A^\bullet) \\ \uparrow{\scriptstyle t} & & \uparrow{\scriptstyle t_0} \\ \text{Ext}^{-2}_A(A, \omega_A^\bullet) & \longleftarrow & \text{Ext}^{-2}_A(A/fA, \omega_A^\bullet) \end{array}$$

It follows that the length of the image of the top horizontal map is at most the length of $\text{Ext}^{-2}_A(A/fA, \omega_A^\bullet)$ plus the length of the cokernel of t_0.

However, if we replace $\mathcal{L}$ by $\mathcal{L}^n$ for $n > 1$, then we can use

$$t^n : M_n = \Gamma(U, \mathcal{L}^{\otimes n}) \longrightarrow \Gamma(U, \mathcal{O}_U) = A$$

instead of t. This replaces t_0 by its nth power. Thus the length of the image of the map $\text{Ext}^{-2}_A(M_n, \omega_A^\bullet) \leftarrow \text{Ext}^{-2}_A(M_{n,0}, \omega_A^\bullet)$ is at most the length of $\text{Ext}^{-2}_A(A/fA, \omega_A^\bullet)$ plus the length of the cokernel of

$$t_0^n : \text{Ext}^{-2}_A(A/fA, \omega_A^\bullet) \longrightarrow \text{Ext}^{-2}_A(M_{n,0}, \omega_A^\bullet)$$

Via the isomorphism $M_0 \cong A/fA$ the map t_0 becomes $g : A/fA \to A/fA$ for some $g \in A/fA$ and via the corresponding isomorphisms $M_{n,0} \cong A/fA$ the map t_0^n becomes $g^n : A/fA \to A/fA$. Thus the length of the cokernel above is the length of the quotient of $\mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet)$ by g^n. Since $\mathrm{Ext}_A^{-2}(A/fA, \omega_A^\bullet)$ is a finite A-module with support T of dimension 1 and since $V(g) \cap T$ consists of the closed point by our choice of t this length grows linearly in n by Algebra, Lemma 00L8.

To finish the proof we prove the final assertion. Assume $f \in \mathfrak{m} \subset A$ satisfies (1), (2), (3), A is (S_2), and $\dim(A) \geq 4$. Condition (1) implies A is catenary, see Dualizing Complexes, Lemma 0A80. Then $\mathrm{Spec}(A)$ is equidimensional by Local Cohomology, Lemma 0FIW. Thus $\dim(A_\mathfrak{p}) + \dim(A/\mathfrak{p}) \geq 4$ for every prime $\mathfrak{p}$ of A. Then $\mathrm{depth}(A_\mathfrak{p}) \geq \min(2, \dim(A_\mathfrak{p})) \geq \min(2, 4 - \dim(A/\mathfrak{p}))$ and hence (4) holds. □

References

[1] Kollár, János. *Maps between local Picard groups.* Algebr. Geom. 3 (4), 461–495 (2016).